



پاکستان ایئرپورٹس اتھارٹی

Internship Report

- **Internee Name:** Abdullah Kamran
- **Coordinator:** Abdul Majid
- **Organization:** Pakistan Aviation Authority (PAA)
- **Duration Of Internship:** 23rd June – 24th July 2025
- **Submission Date:** 28th July 2025
- **Project Topic:** Document Management System (DMS)

Project Topic:

Document Management System (DMS)

Internship Overview:

This report outlines my four to five week internship journey at the Pakistan Aviation Authority (PAA) from 23rd June to 24th July 2025, where I developed and deployed a complete Document Management System (DMS) using ASP.NET Core MVC, JavaScript, Entity Framework, and SQL Server under the guidance of Sir Abdul Majid. This internship enhanced my understanding of software development workflows, front-end and back-end technologies, and professional coding practices.

DAILY WORK SUMMARY:

Week 1: Foundation Setup:

S. NO.	DAYS	DATE	DESCRIPTION
1	Monday	23/06/2025	Interview session and setup instructions. Installed Visual Studio and SQL Server Management Studio.
2	Tuesday	24/06/2025	Faced internet access issues. Learned ASP.NET basics and installed .NET 9.0 SDK.
3	Wednesday	25/06/2025	Watched ASP.NET Bootcamp tutorials. Finalized SQL Server Management Studio installation.
4	Thursday	26/06/2025	Learned C# basics and SQL. Developed LocationController and views (Index, Create, and Update).

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Visual Studio 2022

Open recent

▲ This week

DMS.sln
C:\Users\Abdullah Kamran\Downloads\DMS 22/07/2025 11:12 am

▲ This month

paa-main
C:\Users\Abdullah Kamran\Downloads 04/07/2025 12:48 pm

▲ Older

webApp.sln
C:\Users\Abdullah Kamran\Documents\webApp 26/06/2025 3:02 pm

MyWebApp.sln
C:\Users\Abdullah Kamran\source\repos\MyWebApp 24/06/2025 1:26 pm

Get started

- Clone a repository**
Get code from an online repository like GitHub or Azure DevOps
- Open a project or solution**
Open a local Visual Studio project or .sln file
- Open a local folder**
Navigate and edit code within any folder
- Create a new project**
Choose a project template with code scaffolding to get started

[Continue without code →](#)

Step 2 - Determine which version of SQL Server Management Studio to install

Decide which version of SSMS to install. The most common options are:

- The latest release of SQL Server Management Studio 21 that is hosted on Microsoft servers. To install this version, select the following link. The installer downloads a small *bootstrapper* to your *Downloads* folder.

[Download SSMS 21](#)

- If you already have SQL Server Management Studio 21 installed, you can [install another version alongside it](#).
- You can download a bootstrapper or installer for a specific version from the [Release history](#) page and use it to install another version of SSMS.

asp | Clear all

All languages All platforms All project types

- ASP.NET Core Web App (Razor Pages)**
A project template for creating an ASP.NET Core application with example ASP.NET Core Razor Pages content
C# Linux macOS Windows Cloud Service Web
- ASP.NET Core Web API**
A project template for creating a RESTful Web API using ASP.NET Core controllers or minimal APIs, with optional support for OpenAPI and authentication.
C# Linux macOS Windows API Cloud Service Web Web API
- ASP.NET Core Web API (native AOT)**
A project template for creating a RESTful Web API using ASP.NET Core minimal APIs published as native AOT.
C# Linux macOS Windows API Cloud Service Web Web API
- ASP.NET Core gRPC Service**
A project template for creating a gRPC service using ASP.NET Core, with optional support for publishing as native AOT.
C# Linux macOS Windows Cloud Service Web
- ASP.NET Core Empty**
An empty project template for creating an ASP.NET Core application. This template does not have any content in it.
C# Linux macOS Windows Cloud Service Web
- ASP.NET Core Web App (Model-View-Controller)**
A project template for creating an ASP.NET Core application with example ASP.NET Core MVC Views and Controllers. This template can also be used for RESTful HTTP services.
C# Linux macOS Windows Cloud Service Web

Back Next

```
AccountController.cs Login.cshtml _Layout.cshtml Index.cshtml Moist.cshtml Moist.js MOController.cs Moist.js List.cshtml Solution Explorer
DMS
DMS.Controllers.LocationController
Delete(int id)
using DMS.Models;
using DMS.VMs;
using Microsoft.AspNetCore.Mvc;
using Microsoft.EntityFrameworkCore;
using System.Threading;
using static Microsoft.EntityFrameworkCore.DbLoggerCategory;

namespace DMS.Controllers
{
    public class LocationController : Controller
    {
        private readonly EFContext _db;
        // private readonly Database _ef;

        public LocationController(EFContext db)
        {
            _db = db;
        }

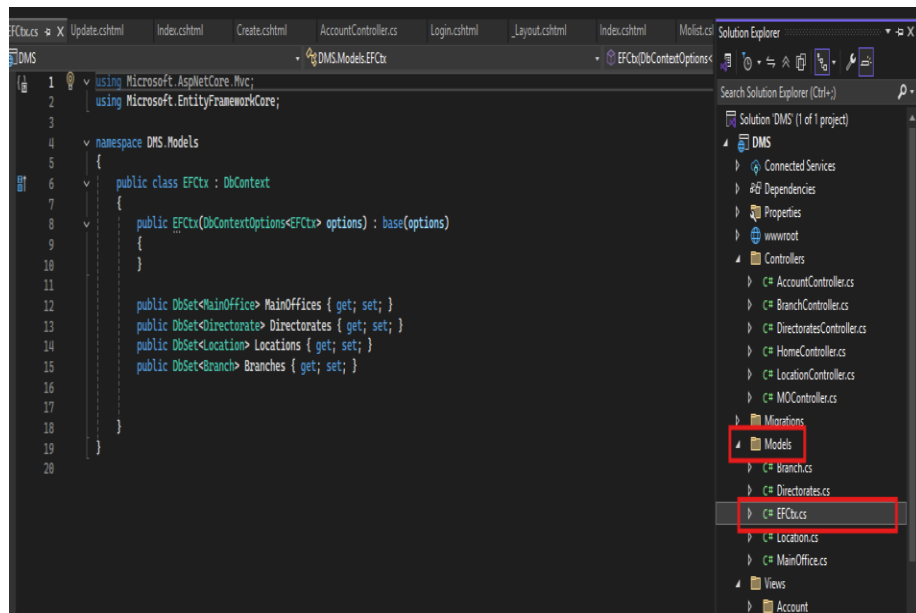
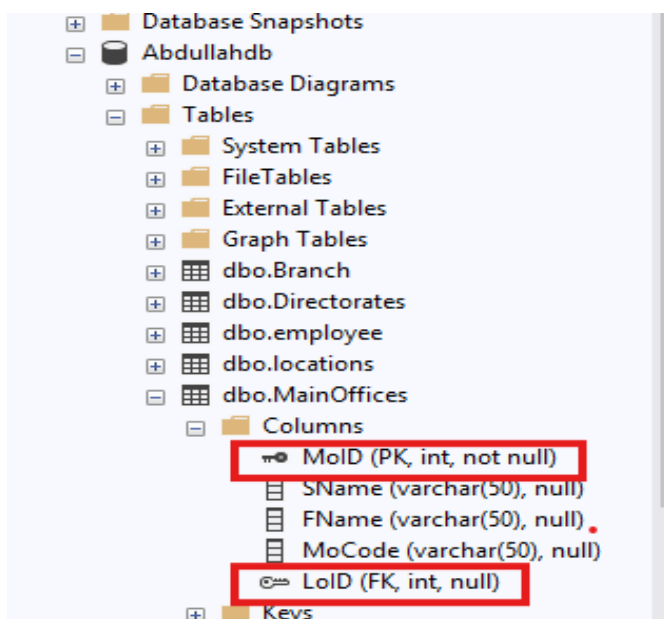
        public IActionResult Index()
        {
            ViewBag.Message = TempData["Message"]?.ToString();
            var locations = _db.Locations.ToList();
            return View(locations);
        }

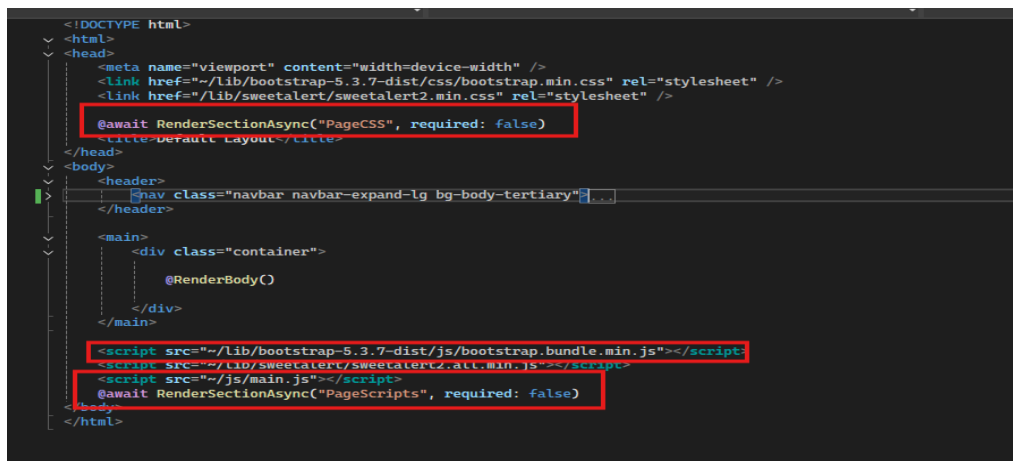
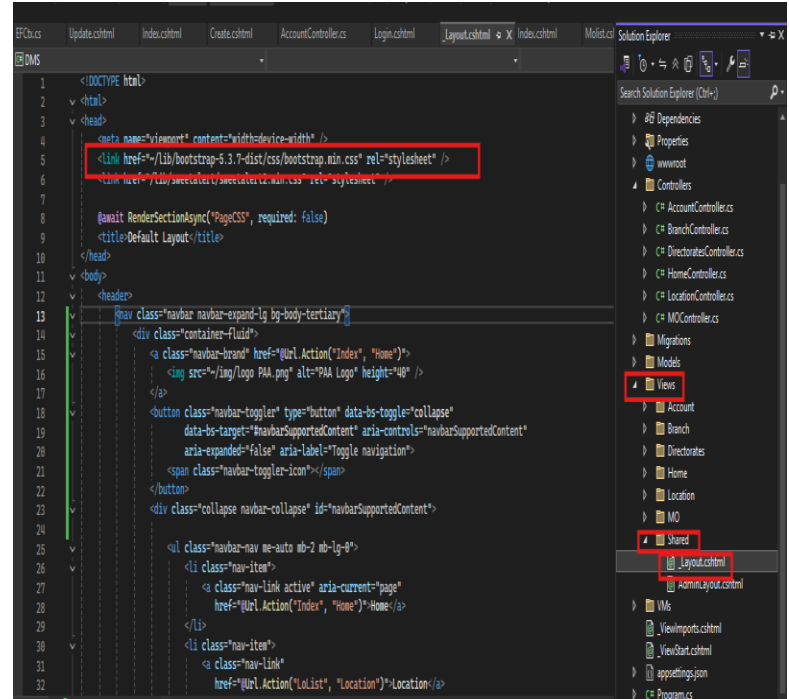
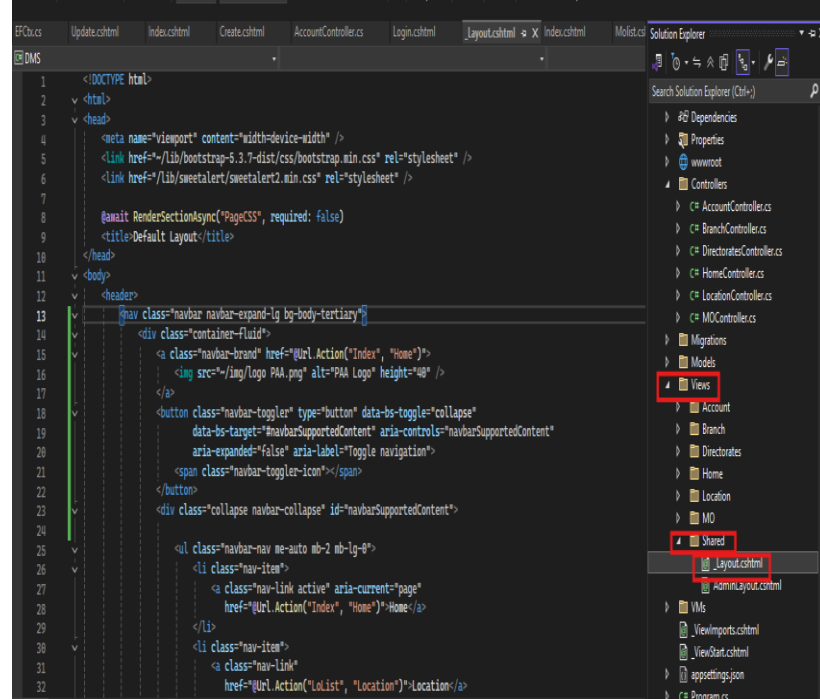
        public IActionResult create()
        {
            return View();
        }
    }
}
```

```
Create.cshtml AccountController.cs Login.cshtml _Layout.cshtml Index.cshtml Moist.cshtml Moist.js MOController.cs List.cshtml Solution Explorer
DMS
<title>Create User</title>
<div class="row">
    <div class="col-12 text-center">
        <h1>Create New User</h1>
    </div>
    <div class="row">
        <div class="col-3">
        </div>
        <div class="col-6">
            <div>
                <div class="input-group mb-3">
                    <span class="input-group-text">Short Name</span>
                    <input type="text" class="form-control" id="txtName">
                </div>
                <div class="input-group mb-3">
                    <span class="input-group-text">Full Name</span>
                    <input type="text" class="form-control" id="txtName">
                </div>
                <div class="input-group mb-3">
                    <span class="input-group-text">Code</span>
                    <input type="text" class="form-control" id="txtCode">
                </div>
                <input type="button" value="Save" id="btnSave" class="btn btn-success">
            </div>
        </div>
    </div>
</div>
```

❖ Week 2: Building Database and MVC Structure:

S.NO	DAYS	DATES	DESCRIPTION
1	Monday	30/06/25	Studied Primary and Foreign Keys. Built MainOfficeController and views.
2	Tuesday	01/07/25	Added Directorate and Branch modules. Implemented Entity Framework Core.
3	Wednesday	02/07/25	Integrated layout and styles. Created models.
4	Thursday	03/07/25	Imported Bootstrap. Reviewed Admin layout.
5	Friday	04/07/25	Added JavaScript libraries. Studied JS syntax and DOM.





❖ Week 3: JavaScript & APIs

S.NO.	DAYS	DATE	DESCRIPTION
1	Monday	07/07/25	JavaScript functions and DOM events.
2	Tuesday	08/07/25	Learning how to fetch an API from JavaScript.
3	Wednesday	09/07/25	Learned Fetch API and used in project.
4	Friday	11/07/25	Continue learned Fetch API and used in project.

```
<!DOCTYPE html>
<html>
<head>
<meta name="viewport" content="width=device-width" />
<link href="/Lib/bootstrap-5.3.7-dist/css/bootstrap.min.css" rel="stylesheet" />
<link href="/Lib/sweetalert/sweetalert2.min.css" rel="stylesheet" />
@await RenderSectionAsync("PageCSS", required: false)
</head>
<body>
<header>
<nav class="navbar navbar-expand-lg bg-body-tertiary">
</header>
<main>
<div class="container">
@RenderBody()
</div>
</main>
<script src="/Lib/bootstrap-5.3.7-dist/js/bootstrap.bundle.min.js"></script>
<script src="/Lib/sweetalert/sweetalert2.all.min.js"></script>
<script src="/js/main.js"></script>
@await RenderSectionAsync("PageScripts", required: false)
</body>
</html>
```

Locations's List

Code	Short Name	Full Name	Edit & Delete
HQ	HQs	Headquarter	Edit Delete
BKIA	BKIA	Bacha Khan International airport	Edit Delete
OPIS	IIAP	Islamabad	Edit Delete
PAA	PAA	Pakistan Aviation Authority	Edit Delete
KUU	KU	Karachi University	Edit Delete
pak	HAKP	Pakistan Aviation Authority	Edit Delete
JIA	JIA	Jinnah International Airport	Edit Delete
INGIA	GIA	Gwadar International Airport	Edit Delete
MMIAAP	MIAP	Multan International Airport	Edit Delete
IIAAPP	IIAP	Islamabad International Airport	Edit Delete
QOQT	QIAP	Quaid-e-Azam International Airport	Edit Delete

[Create New User](#)

```
const abc = async () => {
  try {
    const res = await fetch('/location/getData');
    if (res.ok) {
      const isJson = res.headers.get('content-type')?.includes('application/json');
      let data = isJson && await res.json();
      return data;
    }
    else {
      iziToast.error({
        message: res.statusText,
      });
    }
  }
  catch (error) {
    iziToast.error({
      message: error,
    });
  }
  return null;
}

ready(function () {
  let tbl = document.getElementById('tbl_load');
  let tbody = tbl.getElementsByTagName('tbody');
  if (tbody && tbody.length > 0) {
    tbl.removeChild(tbody[0]);
    tbody = tbl.createTbody();
  }
}
```

Create New User

Short Name

Full Name

Code

Save

[Back to List](#)

❖ Week 4: (Fetch & JS):

S. NO.	DAYS	DATE	DESCRIPTION
1	Monday	14/07/25	Created lolist.js for Location. Connected with controller.
2	Tuesday	15/07/25	Developed molist.js and save.js.
3	Wednesday	16/07/25	Implemented MO save functionality.
4	Thursday	17/07/25	Finalized fetch and JS logic.

```

public IActionResult LoList()
{
    return View();
}

[HttpGet]
public IActionResult getLoData()
{
    var abc = _db.Locations.ToList();
    return Json(new { xyz = abc });
}

[HttpPost]
public IActionResult submitupdate([FromBody] LocVM lo)
{
    try
    {
        var existingLocation = _db.Locations.FirstOrDefault(x => x.LoID == lo.LoID);
        if (existingLocation != null)
        {
            existingLocation.ShortName = lo.SNm;
            existingLocation.FullName = lo.FNm;
            existingLocation.LoCode = lo.LoCd;

            _db.SaveChanges();
            return Json(new { success = true });
        }

        return Json(new { success = false, message = "Location not found." });
    }
    catch (Exception ex)
    {
        return Json(new { success = false, message = ex.Message });
    }
}

```

```

public IActionResult Update(int id)
{
    var office = _db.MainOffices.FirstOrDefault(m => m.MoID == id);
    if (office == null)
        return NotFound();

    ViewBag.LocationList = _db.Locations.ToList();
    return View(office);
}

public IActionResult MoList()
{
    return View();
}

[HttpGet]
[Route("MO/GetMoData")]
public IActionResult GetMoData()
{
    var mos = _db.MainOffices
        .Include(m => m.Location)
        .ToList()
        .Select(m => new
        {
            m.MoID,
            m.MoCode,
            m.ShortName,
            m.FullName,
            LocationName = m.Location != null ? m.Location.ShortName : "N/A"
        });

    return Json(new { records = mos });
}

```

Home Location Main Office Directorate Branch Dropdown Disabled

Update Record of Location

Short Name HQs

Full Name Headquarter

Code HQ

Update

[Back to List](#)

Home Location Main Office Directorate Branch Dropdown Disabled

Update Record of Main Office

Location HQs (HQ)

-- Select Location --

HQs (HQ)

BKIAI (BKA)

IIAP (OPIS)

PAA (PAA)

KU (KUU)

hAKP (pak)

JIA (JIAA)

GIA (NGIA)

MIAP (MMIAAP)

IIAP (IIAAP)

QIAP (QPQT)

Update

[Back to List](#)

```

1 //ready(function () {
2   let bta = document.getElementById('btupdate');
3
4   bta.addEventListener('click', function () {
5
6       (async () => {
7           let lid = document.getElementById('trid').value;
8           let snm = document.getElementById('trishm').value;
9           let fnm = document.getElementById('trifnm').value;
10          let code = document.getElementById('tricode').value;
11          let dcl = document.getElementById('dcllocation').value;
12
13          let lo = {};
14          lo.LoID = lid;
15          lo.SNm = snm;
16          lo.FNm = fnm;
17          lo.LoCd = code;
18          lo.LoId = dcl;
19
20          const response = await fetch('/MO/submitupdate', {
21              method: 'POST',
22              headers: {
23                  'Content-Type': 'application/json'
24              },
25              body: JSON.stringify(lo)
26          });
27
28          if (response.ok) {
29              console.log('success headers', response.headers.get('content-type'));
30              const data = isJson ? await response.json() : null;
31              if (data && data.success === true) {
32                  Swal.fire({
33                      title: 'Aksiya Lagi! Do you want to save the changes?',
34                      showCancelButton: true,
35                  });
36              }
37          }
38      })();
39   });
40 })

```

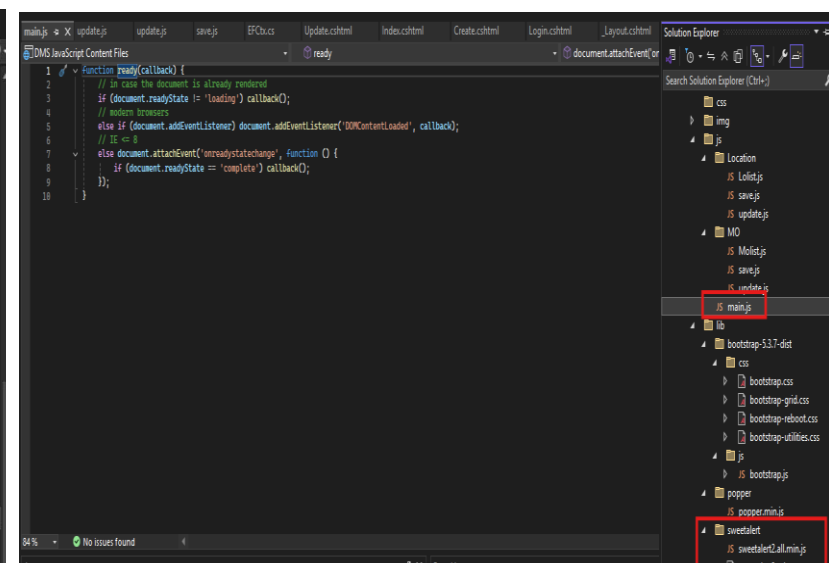
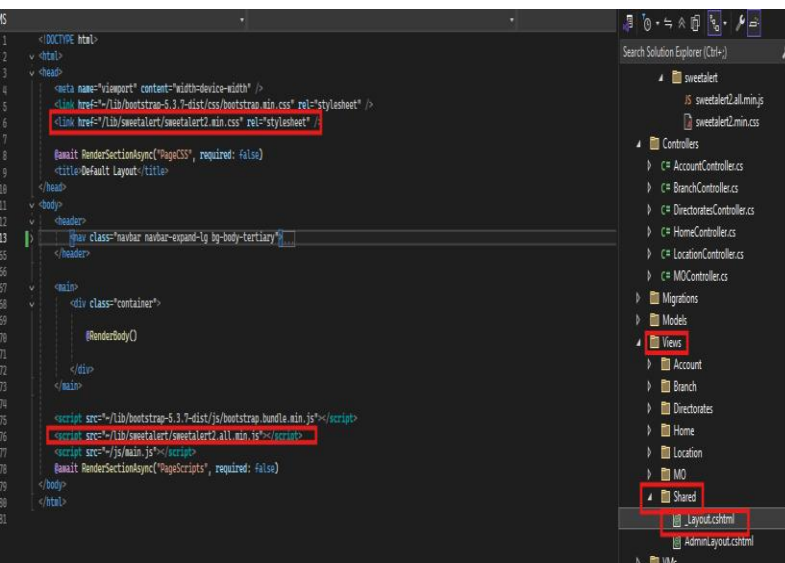
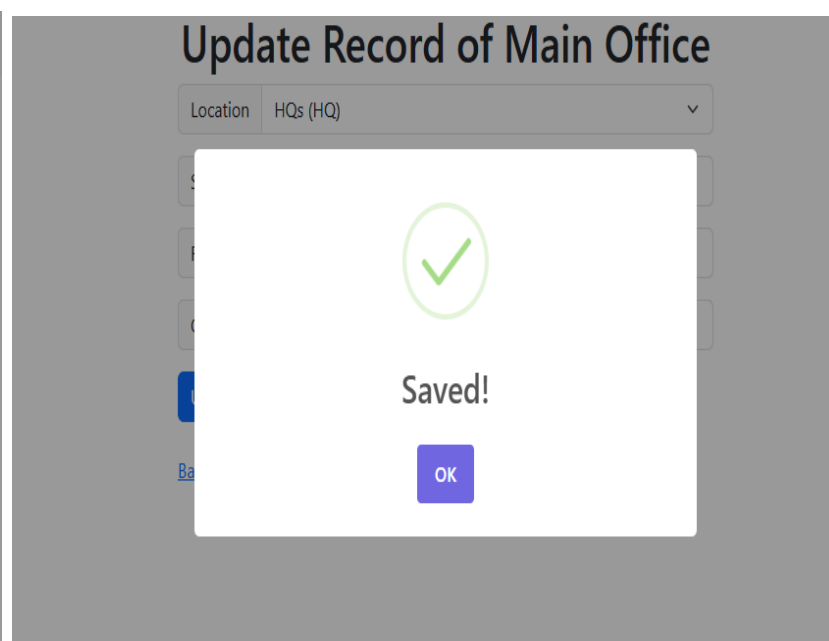
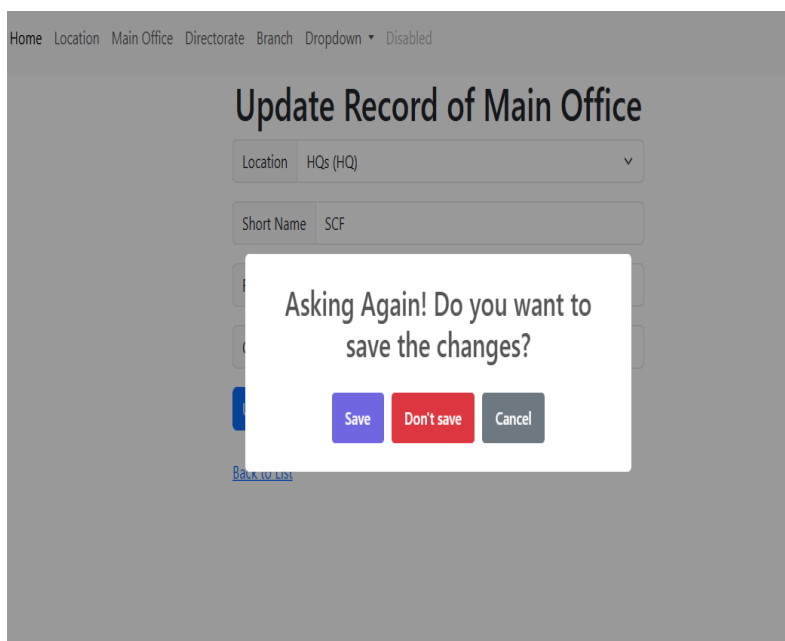
```

1 //ready(function () {
2   let bta = document.getElementById('btupdate');
3
4   bta.addEventListener('click', function () {
5
6       (async () => {
7           let snm = document.getElementById('trishm').value;
8           let fnm = document.getElementById('trifnm').value;
9           let code = document.getElementById('tricode').value;
10          let lid = document.getElementById('dcllocation').value;
11
12          let mo = {};
13          mo.MoID = snm;
14          mo.FNm = fnm;
15          mo.LoCd = code;
16          mo.LoId = lid;
17
18          const response = await fetch('/MO/save', {
19              method: 'POST',
20              headers: {
21                  'Content-Type': 'application/json'
22              },
23              body: JSON.stringify(mo)
24          });
25
26          if (response.ok) {
27              console.log('success headers', response.headers.get('content-type'));
28              const data = isJson ? await response.json() : null;
29              if (data && data.success === true) {
30                  Swal.fire({
31                      title: 'Aksiya Lagi! Do you want to save the changes?',
32                      showCancelButton: true,
33                  });
34              }
35          }
36      })();
37   });
38 })

```

❖ Week 5: Alerts, Deletion, and Login Integration:

S. NO.	DAYS	DATE	DESCRIPTION
1	Monday	21/07/25	Integrated SweetAlert and iziToast
2	Tuesday	22/07/25	Customized alert messages in CRUD
3	Wednesday	23/07/25	Added delete functionality in controllers
4	Thursday	24/07/25	: Designed homepage and login UI. Created AccountController





DMS Project - Document Management System

by Abdullah Kamran | Coordinator: Abdul Majid

Login

Username

Password

Sign In



DMS PROJECT

Document Management System

Submitted By:
Abdullah Kamran

Coordinator:
Abdul Majid

Institute:
Pakistan Airports Authority (PAA)



Locations's List

Code	Short Name	Full Name	Edit & Delete
HQ	HQs	Headquarter	Edit Delete
BKA	BKIAP	Bacha khan International airport	Edit Delete
OPIS	IIAP	Islamabad	Edit Delete
PAA	PAA	Pakistan Aviation Authority	Edit Delete
KUU	KU	Karachi University	Edit Delete
pak	hAKP	Pakistan Aviation Authority	Edit Delete
JJAA	JIA	Jinnah International Airport	Edit Delete
NGIA	GIA	Gwadar International Airport	Edit Delete
MMIAAP	MIAP	Multan International Airport	Edit Delete
IIAAPP	IIAP	Islamabad International Airport	Edit Delete
QPQT	QIAP	Quaid-e-Azam International Airport	Edit Delete



Update Record of Location

Short Name	HQs
Full Name	Headquarter
Code	HQ
<button>Update</button>	

[Back to List](#)



Create New User

Short Name	HQs
Full Name	Pakistan Aviation Authority
Code	PCAA
<button>Save</button>	

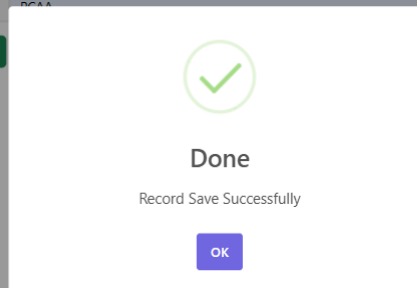
[Back to List](#)



Create New User

Short Name	HQs
Full Name	Pakistan Aviation Authority
Code	PCAA
<button>Save</button>	

[Back to List](#)



Main Office's List

Code	Short Name	Full Name	Location Name	Edit & Delete
SC	SCF	Support & Corporate Functions	HQs	Edit Delete
def	HQs	Airprort Management	hAKP	Edit Delete
PCAA	PAA	Pakistan Aviation Authority	BKIAP	Edit Delete
KPK	KPK	Peshawar, Khyber Pakhtunkhwa	BKIAP	Edit Delete
ISB	ISB	Islamabad, Punjab	IIAP	Edit Delete
Khi	Sindh	Karachi (Main HQ of PAA)	PAA	Edit Delete
Khi	Sindh	Karachi, Sindh	KU	Edit Delete
Gwdr	Bal	Gwadar, Balochistan	GIA	Edit Delete
Khi	QJP	Karachi (Former name of Jinnah International)	GIA	Edit Delete

[Create New Office](#)

Create New Main Office

Location

-- Select Location --

Short Name

-- Select Location --

Full Name

HQs (HQ)

Code

BKIAP (BKA)

Save

IIAP (OPIS)

CU (KUU)

PAA (PAA)

hAKP (pak)

KU (KUU)

JIA (JIAP)

GIA (NGIA)

MIAP (MMIAAP)

IIAP (IIAAPP)

QIAP (QPQT)

HQs (PCAA)

[Back to List](#)

Update Record of Main Office

Location

HQs (HQ)

Short Name

-- Select Location --

Full Name

HQs (HQ)

Code

BKIAP (BKA)

Update

IIAP (OPIS)

CU (KUU)

PAA (PAA)

hAKP (pak)

KU (KUU)

JIA (JIAP)

GIA (NGIA)

MIAP (MMIAAP)

IIAP (IIAAPP)

QIAP (QPQT)

HQs (PCAA)

[Back to List](#)

Update Record of Main Office

Location

HQs (HQ)

Short Name

PAA

Full Name

Pakistan Aviation Authority

Code

PCAA

[Update](#)

[Back to List](#)

Asking Again! Do you want to save the changes?

Save

Don't save

Cancel

❖ **Skills Acquired:**

- ASP.NET Core MVC architecture
- Entity Framework Core
- SQL Server Management Studio (SSMS)
- JavaScript (Vanilla, Fetch API)
- Bootstrap & Responsive UI
- SweetAlert and iziToast integration
- MVC Folder Structure
- Controller, View, Model pattern
- Authentication logic with sessions

Conclusion

This internship has significantly strengthened my web development capabilities and understanding of real-world software projects. Under the mentorship of Sir Abdul Majid, I successfully built a complete Document Management System (DMS) using modern frameworks. It gave me confidence in my backend logic, front-end interaction, and full-stack development practices.

I am thankful to PAA and the entire technical team for this valuable opportunity.

